OPERATING SYSYTEMS

1. What is the primary objective of process scheduling in an operating system?
A) Cache management
B) Linked file allocation
C) Thread management
D) Context switching

2. Which scenario best illustrates the concept of CPU scheduling?
A) Virtual memory management
B) Page replacement
C) Priority scheduling
D) Deadlock detection

3. What problem does context switching mainly attempt to solve?
A) Linked file allocation
B) Priority scheduling
C) Resource allocation
D) CPU utilization

4. In modern operating systems, why is process states important?
A) Interrupt handling
B) Resource allocation
C) Mutex locking
D) Indexed file allocation

5. Which system limitation led to the development of inter-process communication?
A) Device management
B) Round robin scheduling
C) Shortest job first scheduling
D) Process scheduling

6. What is the main benefit achieved through deadlock detection?
A) Process synchronization
B) Context switching
C) Disk scheduling
D) Critical section control

7. Which system behavior most likely requires deadlock prevention?
A) Disk scheduling
B) Deadlock detection

C) Resource allocation
D) Round robin scheduling

8. What challenge in resource management is addressed by deadlock avoidance?
A) Round robin scheduling
B) Page replacement
C) Critical section control
D) Process synchronization

9. Which of the following best describes the purpose of Banker's algorithm?
A) Mutex locking
B) Context switching
C) Cache management
D) Round robin scheduling

10. What system performance issue can be reduced by using race condition?
A) Indexed file allocation
B) Thread management
C) Mutex locking
D) Page replacement

11. What is the primary objective of critical section in an operating system?
A) Context switching
B) Indexed file allocation
C) Virtual memory management
D) Priority scheduling

12. Which scenario best illustrates the concept of process synchronization?
A) Context switching
B) Process scheduling
C) Thread management
D) Semaphore synchronization

13. What problem does semaphores mainly attempt to solve?
A) Priority scheduling
B) Deadlock prevention
C) Cache management
D) Shortest job first scheduling

14. In modern operating systems, why is mutex locks important?
A) Semaphore synchronization

B) Disk scheduling
C) Memory management
D) Priority scheduling

15. Which system limitation led to the development of monitors?
A) Memory management
B) Page replacement
C) Mutex locking
D) Deadlock prevention

16. What is the main benefit achieved through multithreading?
A) Linked file allocation
B) Inter-process communication
C) Mutex locking
D) Context switching

17. Which system behavior most likely requires kernel threads?
A) Semaphore synchronization
B) Mutex locking
C) Cache management
D) Indexed file allocation

18. What challenge in resource management is addressed by user threads?
A) Demand paging
B) Memory management
C) Resource allocation
D) Critical section control

19. Which of the following best describes the purpose of thread scheduling?
A) Mutex locking
B) Thread management
C) Semaphore synchronization
D) Round robin scheduling

20. What system performance issue can be reduced by using virtual memory?
A) Device management
B) CPU utilization
C) Virtual memory management
D) Round robin scheduling

21. What is the primary objective of paging in an operating system?
A) Critical section control
B) Resource allocation
C) Page replacement
D) File system management

22. Which scenario best illustrates the concept of segmentation?
A) Page replacement
B) Process scheduling
C) Context switching
D) Memory management

23. What problem does demand paging mainly attempt to solve?
A) Semaphore synchronization
B) Context switching
C) Disk scheduling
D) Shortest job first scheduling

24. In modern operating systems, why is page replacement important?
A) Deadlock detection
B) Virtual memory management
C) Context switching
D) Semaphore synchronization

25. Which system limitation led to the development of LRU algorithm?
A) Deadlock detection
B) Round robin scheduling
C) File system management
D) Indexed file allocation

26. What is the main benefit achieved through FIFO algorithm?
A) Memory management
B) Cache management
C) Thread management
D) Page replacement

27. Which system behavior most likely requires optimal page replacement?
A) Priority scheduling
B) Critical section control
C) Shortest job first scheduling
D) Linked file allocation

28. What challenge in resource management is addressed by thrashing?
A) Demand paging
B) Resource allocation
C) Semaphore synchronization
D) Priority scheduling

29. Which of the following best describes the purpose of working set model?
A) File system management
B) Thread management
C) Shortest job first scheduling
D) Context switching

30. What system performance issue can be reduced by using memory management?
A) Resource allocation
B) Shortest job first scheduling
C) Semaphore synchronization
D) Thread management

31. What is the primary objective of internal fragmentation in an operating system?
A) Round robin scheduling
B) Linked file allocation
C) Interrupt handling
D) Disk scheduling

32. Which scenario best illustrates the concept of external fragmentation?
A) Resource allocation
B) File system management
C) Shortest job first scheduling
D) Deadlock detection

33. What problem does disk scheduling mainly attempt to solve?
A) File system management
B) CPU utilization
C) Critical section control
D) Page replacement

34. In modern operating systems, why is SCAN algorithm important?
A) Process scheduling
B) Cache management
C) Priority scheduling
D) Inter-process communication

35. Which system limitation led to the development of C-SCAN algorithm?
A) Priority scheduling
B) Linked file allocation
C) Mutex locking
D) Shortest job first scheduling

36. What is the main benefit achieved through SSTF scheduling?
A) Mutex locking
B) Inter-process communication
C) Thread management
D) Semaphore synchronization

37. Which system behavior most likely requires file system?
A) Mutex locking
B) Inter-process communication
C) Shortest job first scheduling
D) Page replacement

38. What challenge in resource management is addressed by file allocation?
A) Semaphore synchronization
B) Indexed file allocation
C) Deadlock detection
D) Resource allocation

39. Which of the following best describes the purpose of indexed allocation?
A) Context switching
B) Resource allocation
C) Deadlock prevention
D) Demand paging

40. What system performance issue can be reduced by using linked allocation?
A) Linked file allocation
B) Semaphore synchronization
C) Page replacement
D) Shortest job first scheduling

41. What is the primary objective of inode in an operating system?
A) Process scheduling
B) Context switching
C) Page replacement
D) Critical section control

42. Which scenario best illustrates the concept of file descriptors?
A) Demand paging
B) Mutex locking
C) Memory management
D) Virtual memory management

43. What problem does I/O buffering mainly attempt to solve?
A) Priority scheduling
B) Inter-process communication
C) File system management
D) Page replacement

44. In modern operating systems, why is spooling important?
A) Interrupt handling
B) Demand paging
C) Disk scheduling
D) Shortest job first scheduling

45. Which system limitation led to the development of interrupts?
A) Resource allocation
B) Round robin scheduling
C) Demand paging
D) Interrupt handling

46. What is the main benefit achieved through system calls?
A) Inter-process communication
B) Process synchronization
C) Mutex locking
D) Demand paging

47. Which system behavior most likely requires kernel mode?
A) Deadlock detection
B) CPU utilization
C) Thread management
D) Resource allocation

48. What challenge in resource management is addressed by user mode?
A) Device management
B) Deadlock detection
C) Linked file allocation
D) Inter-process communication

49. Which of the following best describes the purpose of microkernel?
A) Deadlock prevention
B) Virtual memory management
C) Resource allocation
D) Context switching

50. What system performance issue can be reduced by using monolithic kernel?
A) Mutex locking
B) Page replacement
C) Round robin scheduling
D) Interrupt handling

51. What is the primary objective of hybrid kernel in an operating system?
A) Semaphore synchronization
B) Process scheduling
C) Linked file allocation
D) Resource allocation

52. Which scenario best illustrates the concept of multiprogramming?
A) Context switching
B) Cache management
C) Linked file allocation
D) Deadlock prevention

53. What problem does multitasking mainly attempt to solve?
A) Linked file allocation
B) Deadlock detection
C) Critical section control
D) Inter-process communication

54. In modern operating systems, why is time sharing important?
A) Round robin scheduling
B) CPU utilization
C) Demand paging
D) Device management

55. Which system limitation led to the development of device drivers?
A) Virtual memory management
B) Priority scheduling
C) Shortest job first scheduling
D) Deadlock prevention

56. What is the main benefit achieved through DMA transfer?
A) Mutex locking
B) Round robin scheduling
C) Linked file allocation
D) Critical section control

57. Which system behavior most likely requires cache coherence?
A) Round robin scheduling
B) File system management
C) Shortest job first scheduling
D) Deadlock detection

58. What challenge in resource management is addressed by NUMA architecture?
A) Memory management
B) Critical section control
C) Semaphore synchronization
D) Thread management

59. Which of the following best describes the purpose of distributed OS?
A) Device management
B) Thread management
C) Page replacement
D) Priority scheduling

60. What system performance issue can be reduced by using real-time scheduling?
A) Virtual memory management
B) Mutex locking
C) Process scheduling
D) Priority scheduling

61. What is the primary objective of priority scheduling in an operating system?
A) Priority scheduling
B) Device management
C) Demand paging
D) Virtual memory management

62. Which scenario best illustrates the concept of round robin scheduling?
A) Process synchronization
B) Virtual memory management
C) Indexed file allocation
D) Semaphore synchronization

63. What problem does shortest job first scheduling mainly attempt to solve?
A) Process synchronization
B) CPU utilization
C) Page replacement
D) Resource allocation

64. Which system limitation led to the development of process scheduling?
A) Mutex locking
B) Shortest job first scheduling
C) Page replacement
D) Disk scheduling

65. What is the main benefit achieved through CPU scheduling?
A) Context switching
B) File system management
C) Device management
D) Deadlock detection

66. Which system behavior most likely requires context switching?
A) CPU utilization
B) Memory management
C) File system management
D) Round robin scheduling

67. What challenge in resource management is addressed by process states?
A) Priority scheduling
B) Context switching
C) Cache management
D) Process scheduling

68. Which of the following best describes the purpose of inter-process communication?
A) Resource allocation
B) Page replacement
C) Deadlock prevention
D) Inter-process communication

69. What system performance issue can be reduced by using deadlock detection?
A) Interrupt handling
B) Virtual memory management
C) Linked file allocation
D) CPU utilization

70. What is the primary objective of deadlock prevention in an operating system?
A) Thread management
B) Disk scheduling
C) Demand paging
D) Deadlock detection

71. Which scenario best illustrates the concept of deadlock avoidance?
A) Deadlock prevention
B) Device management
C) File system management
D) Round robin scheduling

72. What problem does Banker's algorithm mainly attempt to solve?
A) Thread management
B) Virtual memory management
C) Context switching
D) Cache management

73. In modern operating systems, why is race condition important?
A) Memory management
B) Critical section control
C) File system management
D) Resource allocation

74. Which system limitation led to the development of critical section?
A) Virtual memory management
B) Cache management
C) Round robin scheduling
D) Device management

75. What is the main benefit achieved through process synchronization?
A) Resource allocation
B) Priority scheduling
C) Memory management
D) CPU utilization

76. Which system behavior most likely requires semaphores?
A) Round robin scheduling
B) Process scheduling
C) Critical section control
D) Mutex locking

77. What challenge in resource management is addressed by mutex locks?
A) Virtual memory management
B) Device management
C) Process scheduling
D) Context switching

78. Which of the following best describes the purpose of monitors?
A) Linked file allocation
B) Device management
C) Demand paging
D) Mutex locking

79. What system performance issue can be reduced by using multithreading?
A) Mutex locking
B) Priority scheduling
C) Inter-process communication
D) Cache management

80. What is the primary objective of kernel threads in an operating system?
A) Deadlock prevention
B) Indexed file allocation
C) Interrupt handling
D) Page replacement

81. Which scenario best illustrates the concept of user threads?
A) Cache management
B) File system management
C) Thread management
D) Priority scheduling

82. What problem does thread scheduling mainly attempt to solve?
A) Deadlock detection
B) Thread management
C) Round robin scheduling
D) Disk scheduling

83. In modern operating systems, why is virtual memory important?
A) Memory management
B) Deadlock detection
C) CPU utilization
D) Virtual memory management

84. Which system limitation led to the development of paging?
A) Inter-process communication
B) Process scheduling
C) File system management
D) Deadlock prevention

85. What is the main benefit achieved through segmentation?
A) Round robin scheduling
B) Priority scheduling
C) Resource allocation
D) File system management

86. Which system behavior most likely requires demand paging?
A) Memory management
B) Demand paging
C) Semaphore synchronization
D) Context switching

87. What challenge in resource management is addressed by page replacement?
A) Indexed file allocation
B) Interrupt handling
C) Disk scheduling
D) Virtual memory management

88. Which of the following best describes the purpose of LRU algorithm?
A) Interrupt handling
B) Critical section control
C) Inter-process communication
D) Linked file allocation

89. What system performance issue can be reduced by using FIFO algorithm?
A) Context switching
B) Cache management
C) Critical section control
D) Mutex locking

90. What is the primary objective of optimal page replacement in an operating system?
A) Context switching
B) Inter-process communication
C) Memory management
D) Device management

91. Which scenario best illustrates the concept of thrashing?
A) Memory management
B) File system management
C) Priority scheduling
D) Round robin scheduling

92. What problem does working set model mainly attempt to solve?
A) Device management
B) Thread management
C) Process synchronization
D) Round robin scheduling

93. In modern operating systems, why is memory management important?
A) Deadlock prevention
B) Thread management
C) Memory management
D) CPU utilization

94. Which system limitation led to the development of internal fragmentation?
A) Interrupt handling
B) Semaphore synchronization
C) Thread management
D) Inter-process communication

95. What is the main benefit achieved through external fragmentation?
A) Round robin scheduling
B) Interrupt handling
C) Virtual memory management
D) Memory management

96. Which system behavior most likely requires disk scheduling?
A) Priority scheduling
B) Resource allocation
C) Thread management
D) Process scheduling

97. What challenge in resource management is addressed by SCAN algorithm?
A) Thread management
B) Virtual memory management
C) File system management
D) Semaphore synchronization

98. Which of the following best describes the purpose of C-SCAN algorithm?
A) Interrupt handling
B) Critical section control
C) Round robin scheduling
D) Deadlock detection

99. What system performance issue can be reduced by using SSTF scheduling?
A) Process scheduling
B) Inter-process communication
C) Critical section control
D) Priority scheduling

100. What is the primary objective of file system in an operating system?
A) Deadlock detection
B) Process synchronization
C) Critical section control
D) Round robin scheduling

101. Which scenario best illustrates the concept of file allocation?
A) Demand paging
B) Process synchronization
C) Thread management
D) Priority scheduling

102. What problem does indexed allocation mainly attempt to solve?
A) Deadlock detection
B) Shortest job first scheduling
C) Mutex locking
D) Context switching

103. In modern operating systems, why is linked allocation important?
A) Process synchronization
B) Process scheduling
C) Semaphore synchronization
D) Deadlock detection

104. Which system limitation led to the development of inode?
A) Process scheduling
B) Deadlock prevention
C) Mutex locking
D) Cache management

105. What is the main benefit achieved through file descriptors?
A) Shortest job first scheduling
B) Resource allocation
C) Round robin scheduling
D) Deadlock detection

106. Which system behavior most likely requires I/O buffering?
A) Priority scheduling
B) Disk scheduling
C) Demand paging
D) Cache management

107. What challenge in resource management is addressed by spooling?
A) Linked file allocation
B) Process scheduling
C) Page replacement
D) Mutex locking

108. Which of the following best describes the purpose of interrupts?
A) Round robin scheduling
B) Cache management
C) Shortest job first scheduling
D) Mutex locking

109. What system performance issue can be reduced by using system calls?
A) Interrupt handling
B) Virtual memory management
C) Indexed file allocation
D) Disk scheduling

110. What is the primary objective of kernel mode in an operating system?
A) File system management
B) Disk scheduling
C) Thread management
D) Shortest job first scheduling

111. Which scenario best illustrates the concept of user mode?
A) Inter-process communication
B) Disk scheduling
C) Deadlock detection

D) File system management

112. What problem does microkernel mainly attempt to solve?
A) Thread management
B) Device management
C) CPU utilization
D) Context switching

113. In modern operating systems, why is monolithic kernel important?
A) Thread management
B) Memory management
C) Interrupt handling
D) Deadlock detection

114. Which system limitation led to the development of hybrid kernel?
A) Linked file allocation
B) Deadlock detection
C) Round robin scheduling
D) Semaphore synchronization

115. What is the main benefit achieved through multiprogramming?
A) Thread management
B) Priority scheduling
C) CPU utilization
D) Indexed file allocation

116. Which system behavior most likely requires multitasking?
A) Context switching
B) Memory management
C) Device management
D) CPU utilization

117. What challenge in resource management is addressed by time sharing?
A) Process synchronization
B) Deadlock prevention
C) Round robin scheduling
D) Interrupt handling

118. Which of the following best describes the purpose of device drivers?
A) Context switching
B) Critical section control
C) Device management

D) Disk scheduling

119. What system performance issue can be reduced by using DMA transfer?
A) Linked file allocation
B) Round robin scheduling
C) Memory management
D) Virtual memory management

120. What is the primary objective of cache coherence in an operating system?
A) Semaphore synchronization
B) Memory management
C) Inter-process communication
D) Critical section control

121. Which scenario best illustrates the concept of NUMA architecture?
A) Demand paging
B) Page replacement
C) Deadlock prevention
D) CPU utilization

122. What problem does distributed OS mainly attempt to solve?
A) Indexed file allocation
B) CPU utilization
C) File system management
D) Shortest job first scheduling

123. In modern operating systems, why is real-time scheduling important?
A) File system management
B) Demand paging
C) Semaphore synchronization
D) Page replacement

124. Which system limitation led to the development of priority scheduling?
A) Process scheduling
B) Demand paging
C) Critical section control
D) Page replacement

125. What is the main benefit achieved through round robin scheduling?
A) Priority scheduling
B) Resource allocation
C) Linked file allocation

D) Demand paging

126. Which system behavior most likely requires shortest job first scheduling?
A) Cache management
B) Indexed file allocation
C) Semaphore synchronization
D) Process scheduling

127. Which of the following best describes the purpose of process scheduling?
A) Context switching
B) Inter-process communication
C) Deadlock detection
D) File system management

128. What system performance issue can be reduced by using CPU scheduling?
A) Process scheduling
B) Virtual memory management
C) Page replacement
D) CPU utilization

129. What is the primary objective of context switching in an operating system?
A) Thread management
B) Disk scheduling
C) Page replacement
D) Resource allocation

130. Which scenario best illustrates the concept of process states?
A) Interrupt handling
B) Memory management
C) Indexed file allocation
D) Device management

131. What problem does inter-process communication mainly attempt to solve?
A) Virtual memory management
B) Page replacement
C) Indexed file allocation
D) Mutex locking

132. In modern operating systems, why is deadlock detection important?
A) Deadlock prevention
B) Thread management
C) Disk scheduling

D) Shortest job first scheduling

133. Which system limitation led to the development of deadlock prevention?
A) File system management
B) Semaphore synchronization
C) Virtual memory management
D) Indexed file allocation

134. What is the main benefit achieved through deadlock avoidance?
A) Thread management
B) Round robin scheduling
C) Virtual memory management
D) Interrupt handling

135. Which system behavior most likely requires Banker's algorithm?
A) Process synchronization
B) Inter-process communication
C) Priority scheduling
D) Deadlock prevention

136. What challenge in resource management is addressed by race condition?
A) Thread management
B) Linked file allocation
C) Mutex locking
D) Priority scheduling

137. Which of the following best describes the purpose of critical section?
A) Shortest job first scheduling
B) Process scheduling
C) Device management
D) Deadlock prevention

138. What system performance issue can be reduced by using process synchronization?
A) Page replacement
B) File system management
C) Deadlock detection
D) Cache management

139. What is the primary objective of semaphores in an operating system?
A) Priority scheduling
B) Shortest job first scheduling
C) CPU utilization

D) Deadlock detection

140. Which scenario best illustrates the concept of mutex locks?
A) Demand paging
B) Round robin scheduling
C) Semaphore synchronization
D) Cache management

141. What problem does monitors mainly attempt to solve?
A) Critical section control
B) Shortest job first scheduling
C) Memory management
D) Process synchronization

142. In modern operating systems, why is multithreading important?
A) Page replacement
B) Resource allocation
C) Linked file allocation
D) Demand paging

143. Which system limitation led to the development of kernel threads?
A) CPU utilization
B) Shortest job first scheduling
C) Interrupt handling
D) Thread management

144. What is the main benefit achieved through user threads?
A) Cache management
B) Disk scheduling
C) Page replacement
D) Process synchronization

145. Which system behavior most likely requires thread scheduling?
A) Thread management
B) Page replacement
C) Priority scheduling
D) Linked file allocation

146. What challenge in resource management is addressed by virtual memory?
A) CPU utilization
B) Critical section control
C) Demand paging

D) Memory management

147. Which of the following best describes the purpose of paging?
A) Cache management
B) Deadlock detection
C) Critical section control
D) Virtual memory management

148. What system performance issue can be reduced by using segmentation?
A) Shortest job first scheduling
B) Interrupt handling
C) Memory management
D) Virtual memory management

149. What is the primary objective of demand paging in an operating system?
A) Memory management
B) Mutex locking
C) Thread management
D) Disk scheduling

150. Which scenario best illustrates the concept of page replacement?
A) Priority scheduling
B) Interrupt handling
C) Virtual memory management
D) Inter-process communication

151. What problem does LRU algorithm mainly attempt to solve?
A) Cache management
B) Demand paging
C) Deadlock detection
D) Virtual memory management

152. In modern operating systems, why is FIFO algorithm important?
A) Disk scheduling
B) Mutex locking
C) Priority scheduling
D) Device management

153. Which system limitation led to the development of optimal page replacement?
A) Round robin scheduling
B) Inter-process communication
C) Virtual memory management

D) Memory management

154. What is the main benefit achieved through thrashing?
A) Semaphore synchronization
B) Linked file allocation
C) Round robin scheduling
D) Indexed file allocation

155. Which system behavior most likely requires working set model?
A) Device management
B) Disk scheduling
C) Shortest job first scheduling
D) Semaphore synchronization

156. What challenge in resource management is addressed by memory management?
A) Demand paging
B) Device management
C) File system management
D) Mutex locking

157. Which of the following best describes the purpose of internal fragmentation?
A) Interrupt handling
B) Linked file allocation
C) Round robin scheduling
D) Page replacement

158. What system performance issue can be reduced by using external fragmentation?
A) Inter-process communication
B) Device management
C) Memory management
D) Disk scheduling

159. What is the primary objective of disk scheduling in an operating system?
A) Disk scheduling
B) Interrupt handling
C) Resource allocation
D) Mutex locking

160. Which scenario best illustrates the concept of SCAN algorithm?
A) Memory management
B) Semaphore synchronization
C) Inter-process communication

D) Round robin scheduling

161. What problem does C-SCAN algorithm mainly attempt to solve?
A) Resource allocation
B) Context switching
C) Mutex locking
D) CPU utilization

162. In modern operating systems, why is SSTF scheduling important?
A) Semaphore synchronization
B) Cache management
C) Device management
D) Deadlock prevention

163. Which system limitation led to the development of file system?
A) Indexed file allocation
B) Process synchronization
C) Resource allocation
D) Shortest job first scheduling

164. What is the main benefit achieved through file allocation?
A) Memory management
B) Deadlock prevention
C) Priority scheduling
D) Mutex locking

165. Which system behavior most likely requires indexed allocation?
A) Priority scheduling
B) Deadlock detection
C) Critical section control
D) Virtual memory management

166. What challenge in resource management is addressed by linked allocation?
A) Page replacement
B) Mutex locking
C) Deadlock detection
D) Context switching

167. Which of the following best describes the purpose of inode?
A) Context switching
B) Process synchronization

C) Virtual memory management
D) Deadlock detection

168. What system performance issue can be reduced by using file descriptors?
A) Critical section control
B) Process scheduling
C) Interrupt handling
D) Inter-process communication

169. What is the primary objective of I/O buffering in an operating system?
A) Demand paging
B) Deadlock prevention
C) Context switching
D) Indexed file allocation

170. Which scenario best illustrates the concept of spooling?
A) Cache management
B) Deadlock prevention
C) CPU utilization
D) Priority scheduling

171. What problem does interrupts mainly attempt to solve?
A) Interrupt handling
B) Priority scheduling
C) Demand paging
D) Page replacement

172. In modern operating systems, why is system calls important?
A) Round robin scheduling
B) Deadlock detection
C) Demand paging
D) Cache management

173. Which system limitation led to the development of kernel mode?
A) Semaphore synchronization
B) Page replacement
C) Virtual memory management
D) Demand paging

174. What is the main benefit achieved through user mode?
A) Mutex locking
B) Interrupt handling

C) Shortest job first scheduling
D) Round robin scheduling

175. Which system behavior most likely requires microkernel?
A) CPU utilization
B) Linked file allocation
C) Thread management
D) Page replacement

176. What challenge in resource management is addressed by monolithic kernel?
A) Virtual memory management
B) Resource allocation
C) Deadlock prevention
D) Device management

177. Which of the following best describes the purpose of hybrid kernel?
A) Page replacement
B) Cache management
C) Mutex locking
D) Thread management

178. What system performance issue can be reduced by using multiprogramming?
A) File system management
B) Semaphore synchronization
C) Context switching
D) Mutex locking

179. What is the primary objective of multitasking in an operating system?
A) Linked file allocation
B) Process scheduling
C) Indexed file allocation
D) Round robin scheduling

180. Which scenario best illustrates the concept of time sharing?
A) Virtual memory management
B) Shortest job first scheduling
C) Round robin scheduling
D) Device management

181. What problem does device drivers mainly attempt to solve?
A) Cache management
B) Linked file allocation

C) Thread management
D) Critical section control

182. In modern operating systems, why is DMA transfer important?
A) File system management
B) Page replacement
C) CPU utilization
D) Device management

183. Which system limitation led to the development of cache coherence?
A) Demand paging
B) Resource allocation
C) Context switching
D) Inter-process communication

184. What is the main benefit achieved through NUMA architecture?
A) Semaphore synchronization
B) Inter-process communication
C) Virtual memory management
D) Deadlock prevention

185. Which system behavior most likely requires distributed OS?
A) Disk scheduling
B) Critical section control
C) Device management
D) Process synchronization

186. What challenge in resource management is addressed by real-time scheduling?
A) Device management
B) Priority scheduling
C) Critical section control
D) Process scheduling

187. Which of the following best describes the purpose of priority scheduling?
A) Deadlock prevention
B) Page replacement
C) Process scheduling
D) Process synchronization

188. What system performance issue can be reduced by using round robin scheduling?
A) Critical section control
B) Inter-process communication

C) Memory management
D) Disk scheduling

189. What is the primary objective of shortest job first scheduling in an operating system?
A) Critical section control
B) Context switching
C) Page replacement
D) Deadlock detection

190. What problem does process scheduling mainly attempt to solve?
A) Deadlock prevention
B) CPU utilization
C) Disk scheduling
D) Inter-process communication

191. In modern operating systems, why is CPU scheduling important?
A) Priority scheduling
B) Critical section control
C) Resource allocation
D) Page replacement

192. Which system limitation led to the development of context switching?
A) Mutex locking
B) Page replacement
C) Shortest job first scheduling
D) Process scheduling

193. What is the main benefit achieved through process states?
A) File system management
B) Memory management
C) Interrupt handling
D) Page replacement

194. Which system behavior most likely requires inter-process communication?
A) Demand paging
B) Page replacement
C) Deadlock detection
D) CPU utilization

195. What challenge in resource management is addressed by deadlock detection?
A) Semaphore synchronization
B) Resource allocation

C) Shortest job first scheduling
D) Page replacement

196. Which of the following best describes the purpose of deadlock prevention?
A) CPU utilization
B) Semaphore synchronization
C) Disk scheduling
D) Memory management

197. What system performance issue can be reduced by using deadlock avoidance?
A) Interrupt handling
B) File system management
C) Context switching
D) Indexed file allocation

198. What is the primary objective of Banker's algorithm in an operating system?
A) Indexed file allocation
B) Virtual memory management
C) Round robin scheduling
D) Device management

199. Which scenario best illustrates the concept of race condition?
A) Process scheduling
B) Process synchronization
C) Context switching
D) Critical section control

200. What problem does critical section mainly attempt to solve?
A) Critical section control
B) Interrupt handling
C) Process scheduling
D) Inter-process communication

201. In modern operating systems, why is process synchronization important?
A) File system management
B) Thread management
C) Deadlock prevention
D) Inter-process communication

202. Which system limitation led to the development of semaphores?
A) Deadlock detection
B) Linked file allocation

C) Process synchronization
D) Page replacement

203. What is the main benefit achieved through mutex locks?
A) CPU utilization
B) Device management
C) Critical section control
D) Deadlock detection

204. Which system behavior most likely requires monitors?
A) Process synchronization
B) Memory management
C) Round robin scheduling
D) Virtual memory management

205. What challenge in resource management is addressed by multithreading?
A) Inter-process communication
B) Memory management
C) Indexed file allocation
D) Round robin scheduling

206. Which of the following best describes the purpose of kernel threads?
A) File system management
B) Interrupt handling
C) Virtual memory management
D) Resource allocation

207. What system performance issue can be reduced by using user threads?
A) File system management
B) Thread management
C) Indexed file allocation
D) Virtual memory management

208. What is the primary objective of thread scheduling in an operating system?
A) File system management
B) Critical section control
C) Disk scheduling
D) Linked file allocation

209. Which scenario best illustrates the concept of virtual memory?
A) Interrupt handling
B) Process synchronization

C) Inter-process communication
D) Thread management

210. What problem does paging mainly attempt to solve?
A) Process synchronization
B) Process scheduling
C) Critical section control
D) Resource allocation

211. In modern operating systems, why is segmentation important?
A) Virtual memory management
B) Deadlock prevention
C) Process synchronization
D) Thread management

212. Which system limitation led to the development of demand paging?
A) Process synchronization
B) Deadlock prevention
C) Context switching
D) Device management

213. What is the main benefit achieved through page replacement?
A) Round robin scheduling
B) Linked file allocation
C) Semaphore synchronization
D) Priority scheduling

214. Which system behavior most likely requires LRU algorithm?
A) Demand paging
B) Disk scheduling
C) Process scheduling
D) Thread management

215. What challenge in resource management is addressed by FIFO algorithm?
A) Round robin scheduling
B) File system management
C) CPU utilization
D) Cache management

216. Which of the following best describes the purpose of optimal page replacement?
A) Indexed file allocation
B) Interrupt handling

C) Round robin scheduling
D) Cache management

217. What system performance issue can be reduced by using thrashing?
A) Interrupt handling
B) File system management
C) Process scheduling
D) Deadlock prevention

218. What is the primary objective of working set model in an operating system?
A) Page replacement
B) Device management
C) Semaphore synchronization
D) Linked file allocation

219. Which scenario best illustrates the concept of memory management?
A) Process scheduling
B) Resource allocation
C) Indexed file allocation
D) Thread management

220. What problem does internal fragmentation mainly attempt to solve?
A) Inter-process communication
B) Indexed file allocation
C) Round robin scheduling
D) Page replacement

221. In modern operating systems, why is external fragmentation important?
A) Critical section control
B) Page replacement
C) Thread management
D) Semaphore synchronization

222. Which system limitation led to the development of disk scheduling?
A) Context switching
B) Deadlock detection
C) Inter-process communication
D) Memory management

223. What is the main benefit achieved through SCAN algorithm?
A) Deadlock prevention

B) Shortest job first scheduling
C) Resource allocation
D) Thread management

224. Which system behavior most likely requires C-SCAN algorithm?
A) Priority scheduling
B) Inter-process communication
C) Mutex locking
D) Memory management

225. What challenge in resource management is addressed by SSTF scheduling?
A) Process scheduling
B) Interrupt handling
C) Shortest job first scheduling
D) Resource allocation

226. Which of the following best describes the purpose of file system?
A) Demand paging
B) Memory management
C) Process synchronization
D) Shortest job first scheduling

227. What system performance issue can be reduced by using file allocation?
A) Memory management
B) Deadlock detection
C) Cache management
D) Resource allocation

228. What is the primary objective of indexed allocation in an operating system?
A) Deadlock detection
B) Inter-process communication
C) Disk scheduling
D) Context switching

229. Which scenario best illustrates the concept of linked allocation?
A) Round robin scheduling
B) File system management
C) Process synchronization
D) Deadlock prevention

230. What problem does inode mainly attempt to solve?
A) Round robin scheduling

B) Process synchronization
C) Shortest job first scheduling
D) Deadlock detection

231. In modern operating systems, why is file descriptors important?
A) Shortest job first scheduling
B) Resource allocation
C) Page replacement
D) Semaphore synchronization

232. Which system limitation led to the development of I/O buffering?
A) Linked file allocation
B) Round robin scheduling
C) Shortest job first scheduling
D) Thread management

233. What is the main benefit achieved through spooling?
A) Resource allocation
B) Deadlock prevention
C) Device management
D) Indexed file allocation

234. Which system behavior most likely requires interrupts?
A) Resource allocation
B) File system management
C) Semaphore synchronization
D) Critical section control

235. What challenge in resource management is addressed by system calls?
A) Deadlock prevention
B) File system management
C) Demand paging
D) Round robin scheduling

236. Which of the following best describes the purpose of kernel mode?
A) Process scheduling
B) Demand paging
C) Priority scheduling
D) File system management

237. What system performance issue can be reduced by using user mode?
A) Resource allocation

B) Thread management
C) Mutex locking
D) Device management

238. What is the primary objective of microkernel in an operating system?
A) Mutex locking
B) Thread management
C) Deadlock detection
D) Shortest job first scheduling

239. Which scenario best illustrates the concept of monolithic kernel?
A) Virtual memory management
B) Inter-process communication
C) File system management
D) CPU utilization

240. What problem does hybrid kernel mainly attempt to solve?
A) Critical section control
B) Deadlock detection
C) Resource allocation
D) Virtual memory management

241. In modern operating systems, why is multiprogramming important?
A) Critical section control
B) Thread management
C) Disk scheduling
D) Mutex locking

242. Which system limitation led to the development of multitasking?
A) Disk scheduling
B) Indexed file allocation
C) Cache management
D) Resource allocation

243. What is the main benefit achieved through time sharing?
A) Deadlock prevention
B) Inter-process communication
C) File system management
D) Context switching

244. Which system behavior most likely requires device drivers?
A) Process scheduling

B) Deadlock detection
C) Mutex locking
D) Inter-process communication

245. What challenge in resource management is addressed by DMA transfer?
A) Virtual memory management
B) Demand paging
C) Cache management
D) File system management

246. Which of the following best describes the purpose of cache coherence?
A) Cache management
B) Context switching
C) Mutex locking
D) Priority scheduling

247. What system performance issue can be reduced by using NUMA architecture?
A) Thread management
B) Deadlock prevention
C) Device management
D) Process synchronization

248. What is the primary objective of distributed OS in an operating system?
A) Deadlock prevention
B) Device management
C) Linked file allocation
D) Disk scheduling

249. Which scenario best illustrates the concept of real-time scheduling?
A) Virtual memory management
B) Indexed file allocation
C) Priority scheduling
D) Demand paging

250. What problem does priority scheduling mainly attempt to solve?
A) Deadlock detection
B) Deadlock prevention
C) Round robin scheduling
D) Page replacement

251. In modern operating systems, why is round robin scheduling important?
A) Resource allocation

B) Cache management
C) File system management
D) Deadlock detection

252. Which system limitation led to the development of shortest job first scheduling?
A) Mutex locking
B) Memory management
C) Indexed file allocation
D) Process synchronization

253. Which system behavior most likely requires process scheduling?
A) Indexed file allocation
B) Inter-process communication
C) Process scheduling
D) Shortest job first scheduling

254. What challenge in resource management is addressed by CPU scheduling?
A) Process synchronization
B) Thread management
C) Indexed file allocation
D) Shortest job first scheduling

255. Which of the following best describes the purpose of context switching?
A) Memory management
B) Demand paging
C) Resource allocation
D) Deadlock detection

256. What system performance issue can be reduced by using process states?
A) Disk scheduling
B) Cache management
C) Indexed file allocation
D) Thread management

257. What is the primary objective of inter-process communication in an operating system?
A) CPU utilization
B) Process scheduling
C) Shortest job first scheduling
D) Semaphore synchronization

258. Which scenario best illustrates the concept of deadlock detection?
A) Process synchronization

B) Device management
C) Page replacement
D) Interrupt handling

259. What problem does deadlock prevention mainly attempt to solve?
A) File system management
B) Virtual memory management
C) Context switching
D) Shortest job first scheduling

260. In modern operating systems, why is deadlock avoidance important?
A) File system management
B) Interrupt handling
C) Mutex locking
D) Semaphore synchronization

261. Which system limitation led to the development of Banker's algorithm?
A) Deadlock prevention
B) Page replacement
C) Round robin scheduling
D) Indexed file allocation

262. What is the main benefit achieved through race condition?
A) Virtual memory management
B) File system management
C) Cache management
D) Mutex locking

263. Which system behavior most likely requires critical section?
A) Deadlock detection
B) Virtual memory management
C) Critical section control
D) Mutex locking

264. What challenge in resource management is addressed by process synchronization?
A) Virtual memory management
B) Process scheduling
C) Deadlock prevention
D) Thread management

265. Which of the following best describes the purpose of semaphores?
A) Semaphore synchronization

B) File system management
C) Device management
D) Thread management

266. What system performance issue can be reduced by using mutex locks?
A) File system management
B) Virtual memory management
C) Inter-process communication
D) Semaphore synchronization

267. What is the primary objective of monitors in an operating system?
A) Memory management
B) Priority scheduling
C) Disk scheduling
D) Cache management

268. Which scenario best illustrates the concept of multithreading?
A) Process synchronization
B) Cache management
C) Resource allocation
D) Device management

269. What problem does kernel threads mainly attempt to solve?
A) Device management
B) Shortest job first scheduling
C) Disk scheduling
D) Page replacement

270. In modern operating systems, why is user threads important?
A) Resource allocation
B) Inter-process communication
C) Deadlock prevention
D) Device management

271. Which system limitation led to the development of thread scheduling?
A) Virtual memory management
B) Disk scheduling
C) Process scheduling
D) Deadlock prevention

272. What is the main benefit achieved through virtual memory?
A) Context switching

B) Deadlock prevention
C) Process synchronization
D) Shortest job first scheduling

273. Which system behavior most likely requires paging?
A) Virtual memory management
B) Shortest job first scheduling
C) Thread management
D) Disk scheduling

274. What challenge in resource management is addressed by segmentation?
A) Priority scheduling
B) Interrupt handling
C) Device management
D) Process synchronization

275. Which of the following best describes the purpose of demand paging?
A) File system management
B) Device management
C) Virtual memory management
D) Inter-process communication

276. What system performance issue can be reduced by using page replacement?
A) Context switching
B) Memory management
C) Interrupt handling
D) Semaphore synchronization

277. What is the primary objective of LRU algorithm in an operating system?
A) Process synchronization
B) Linked file allocation
C) Deadlock prevention
D) Resource allocation

278. Which scenario best illustrates the concept of FIFO algorithm?
A) Inter-process communication
B) Virtual memory management
C) Indexed file allocation
D) Shortest job first scheduling

279. What problem does optimal page replacement mainly attempt to solve?

A) Linked file allocation
B) Demand paging
C) Round robin scheduling
D) Device management

280. In modern operating systems, why is thrashing important?
A) Thread management
B) Page replacement
C) Cache management
D) Process synchronization

281. Which system limitation led to the development of working set model?
A) CPU utilization
B) Virtual memory management
C) Page replacement
D) Critical section control

282. What is the main benefit achieved through memory management?
A) Memory management
B) Disk scheduling
C) Deadlock prevention
D) Demand paging

283. Which system behavior most likely requires internal fragmentation?
A) Semaphore synchronization
B) Virtual memory management
C) Cache management
D) Process synchronization

284. What challenge in resource management is addressed by external fragmentation?
A) Page replacement
B) File system management
C) Interrupt handling
D) Cache management

285. Which of the following best describes the purpose of disk scheduling?
A) Memory management
B) Device management
C) File system management
D) Critical section control

286. What system performance issue can be reduced by using SCAN algorithm?

A) Memory management
B) Process synchronization
C) File system management
D) Device management

287. What is the primary objective of C-SCAN algorithm in an operating system?
A) Priority scheduling
B) Page replacement
C) Round robin scheduling
D) File system management

288. Which scenario best illustrates the concept of SSTF scheduling?
A) Deadlock prevention
B) Shortest job first scheduling
C) Thread management
D) Indexed file allocation

289. What problem does file system mainly attempt to solve?
A) Virtual memory management
B) Deadlock prevention
C) Linked file allocation
D) Interrupt handling

290. In modern operating systems, why is file allocation important?
A) Process synchronization
B) Deadlock prevention
C) Priority scheduling
D) Context switching

291. Which system limitation led to the development of indexed allocation?
A) Critical section control
B) Context switching
C) File system management
D) Thread management

292. What is the main benefit achieved through linked allocation?
A) Thread management
B) Virtual memory management
C) CPU utilization
D) Process scheduling

293. Which system behavior most likely requires inode?

A) Indexed file allocation
B) Process scheduling
C) Critical section control
D) Shortest job first scheduling

294. What challenge in resource management is addressed by file descriptors?
A) Interrupt handling
B) Inter-process communication
C) Mutex locking
D) Context switching

295. Which of the following best describes the purpose of I/O buffering?
A) Process synchronization
B) Shortest job first scheduling
C) Interrupt handling
D) Inter-process communication

296. What system performance issue can be reduced by using spooling?
A) Demand paging
B) Round robin scheduling
C) Semaphore synchronization
D) File system management

297. What is the primary objective of interrupts in an operating system?
A) Deadlock prevention
B) Shortest job first scheduling
C) Memory management
D) Inter-process communication

298. Which scenario best illustrates the concept of system calls?
A) Interrupt handling
B) Deadlock detection
C) Disk scheduling
D) Priority scheduling

299. What problem does kernel mode mainly attempt to solve?
A) Semaphore synchronization
B) Device management
C) Disk scheduling
D) Cache management

300. In modern operating systems, why is user mode important?

A) Critical section control
B) Indexed file allocation
C) Round robin scheduling
D) Virtual memory management

Answer Key

1. B	31. D	61. D	91. A	121. C	151. B	181. A	211. A	241. A	271. D
2. A	32. B	62. B	92. A	122. D	152. C	182. A	212. A	242. C	272. A
3. A	33. A	63. A	93. B	123. A	153. A	183. D	213. B	243. D	273. C
4. B	34. C	64. B	94. C	124. C	154. B	184. D	214. D	244. A	274. B
5. A	35. A	65. C	95. C	125. B	155. A	185. C	215. B	245. C	275. C
6. D	36. B	66. A	96. A	126. B	156. A	186. C	216. C	246. C	276. D
7. C	37. C	67. D	97. D	127. C	157. D	187. C	217. D	247. D	277. D
8. B	38. C	68. C	98. B	128. A	158. D	188. B	218. B	248. A	278. C
9. A	39. B	69. C	99. A	129. C	159. B	189. C	219. B	249. A	279. B
10. A	40. B	70. A	100. D	130. C	160. B	190. C	220. C	250. B	280. B
11. B	41. B	71. A	101. B	131. A	161. D	191. D	221. A	251. B	281. C
12. A	42. A	72. D	102. C	132. B	162. D	192. C	222. D	252. C	282. D
13. D	43. D	73. A	103. B	133. C	163. D	193. C	223. B	253. D	283. A
14. C	44. D	74. D	104. A	134. A	164. A	194. C	224. D	254. A	284. B
15. D	45. C	75. B	105. C	135. B	165. B	195. A	225. A	255. A	285. A
16. A	46. D	76. B	106. B	136. C	166. A	196. B	226. A	256. A	286. C
17. A	47. D	77. B	107. A	137. C	167. D	197. A	227. B	257. D	287. B
18. B	48. D	78. A	108. C	138. C	168. D	198. A	228. A	258. C	288. A
19. C	49. B	79. C	109. B	139. D	169. C	199. D	229. A	259. C	289. C
20. B	50. A	80. D	110. A	140. B	170. B	200. A	230. C	260. B	290. A
21. A	51. D	81. A	111. D	141. C	171. A	201. A	231. C	261. B	291. C
22. C	52. A	82. D	112. D	142. A	172. D	202. A	232. B	262. B	292. A
23. C	53. D	83. A	113. B	143. A	173. C	203. C	233. C	263. C	293. D
24. B	54. C	84. A	114. C	144. D	174. A	204. C	234. D	264. C	294. A
25. D	55. A	85. A	115. A	145. B	175. A	205. C	235. C	265. C	295. D
26. C	56. C	86. D	116. A	146. D	176. D	206. A	236. A	266. D	296. D
27. A	57. D	87. C	117. B	147. A	177. D	207. D	237. D	267. B	297. A
28. C	58. A	88. A	118. A	148. B	178. D	208. A	238. B	268. B	298. D
29. B	59. C	89. A	119. C	149. D	179. A	209. B	239. A	269. B	299. A
30. D	60. C	90. A	120. D	150. D	180. B	210. D	240. B	270. C	300. A